AWS VPC Lab

Security Groups, NACLs and Connectivity Testing

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A step-by-step guide to building a custom VPC, launching two EC2 instances with different security groups, and testing how security group and NACL changes affect traffic.

Lab Overview

Goal: Understand how Security Groups (instance level, stateful) and Network ACLs (subnet level, stateless) control traffic.

What you will build

Resource	Name	Purpose
VPC	lab1-custom	Isolated network with public subnets and an Internet Gateway
Security group	SGA	Allows HTTP and SSH from anywhere
Security group	SGB	Allows HTTP, SSH, and ICMP (ping) from SGA
EC2 instance	Instance-A	Apache web server, uses SGA
EC2 instance	Instance-B	Apache web server, uses SGB

Before you start: an AWS account, a region selected, and a key pair (or use EC2 Instance Connect). Everything fits in the Free Tier (t2.micro) as long as you do not create a NAT gateway.

Step 1: Create the Custom VPC

1. Open the **Amazon VPC Console** and click **Your VPCs → Create VPC**.
2. Select **VPC and more**. This auto-generates subnets, route tables, and an Internet Gateway.
3. Under **Name tag auto-generation**, enter `lab1-custom`.
4. Under **NAT gateways**, select **None**. This lab only needs public subnets.
5. Leave all other settings at their defaults and click **Create VPC**.

Enable auto-assign public IPv4 on the subnets

1. In the VPC console, click **Subnets** in the left menu.
2. Select your new **public** subnets (named `lab1-custom-subnet-public...`).
3. Click **Actions → Edit subnet settings**.
4. Tick **Enable auto-assign public IPv4 address** and click **Save**.

Step 2: Create the Security Groups

Open the **Amazon EC2 Console**, click **Security Groups** in the left sidebar, then **Create security group**. Create SGA first, because SGB refers to it.

Security Group A (SGA)

- **Name:** `SGA`
- **Description:** Security group A for challenge
- **VPC:** `lab1-custom`

Direction	Type	Protocol / Port	Source or Destination
Inbound	HTTP	TCP 80	0.0.0.0/0
Inbound	SSH	TCP 22	0.0.0.0/0
Outbound	All traffic	All	0.0.0.0/0

Click **Create security group**.

Security Group B (SGB)

- **Name:** `SGB`
- **Description:** Security group B for challenge
- **VPC:** `lab1-custom`

Direction	Type	Protocol / Port	Source or Destination
Inbound	HTTP	TCP 80	0.0.0.0/0
Inbound	SSH	TCP 22	0.0.0.0/0
Inbound	All ICMP - IPv4	ICMP	Custom: select **SGA**
Outbound	All traffic	All	0.0.0.0/0

Click **Create security group**.

Step 3: Launch the EC2 Instances

Go to **EC2 → Instances → Launch instances**.

Instance-A

1. **Name:** `Instance-A`
2. **AMI:** Amazon Linux 2023 (or Amazon Linux 2). **Instance type:** t2.micro.
3. **Key pair:** select an existing one or create a new key pair.
4. **Network settings → Edit:** VPC `lab1-custom`; Subnet: any public subnet; Auto-assign public IP: **Enable**; Firewall: **Select existing security group → SGA**.
5. Expand **Advanced details**, scroll to **User data**, and paste this script:

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
```

Then click **Launch instance**.

Instance-B

Repeat every step above with two differences: name it `Instance-B`, and choose **SGB** as the security group.

Rules at a Glance

Keep this table handy. It shows what each change in Step 5 does to the rules.

Item	Starting state	After Change 1	After Change 3
SGA, HTTP 80	0.0.0.0/0	Your home IP /32	Same as Change 1
SGA, SSH 22	0.0.0.0/0	Unchanged	Unchanged
SGB, HTTP 80	0.0.0.0/0	Unchanged	Source = SGA
SGB, ICMP	Source = SGA	Unchanged	Unchanged
Subnet NACL	Default (allow all)	Unchanged	Unchanged

Change 2 adds a **deny** rule to the subnet NACL; see Step 5.

Step 4: Baseline Connectivity Tests

Wait until both instances show **2/2 checks passed**, then run these tests.

#	Test	How	Expected result
1	Web browser	Paste the **public IPv4** of Instance-A (then Instance-B) into a browser tab. Use `http://`, not `https://`.	Apache test page loads
2	EC2 Instance Connect	Select an instance → **Connect → EC2 Instance Connect**	Browser terminal opens on both instances
3	Ping	On Instance-A run `ping <Private-IP-of-Instance-B>`	Replies received (SGB allows ICMP from SGA)
4	Curl	On Instance-B run `curl http://<Public-IP-of-Instance-A>`	Returns the Apache page HTML

Press `Ctrl+C` to stop the ping.

> **Note:** Ping only works from A to B. SGA has no ICMP rule, so pinging A from B will fail. That is expected.

Step 5: Apply Rule Changes and Re-test

Do the changes in order. After each one, run its test and note the result. Undo a change before starting the next if you want to see each effect on its own.

Change 1: Restrict SGA HTTP access to your home IP

1. Go to **Security Groups → SGA → Edit inbound rules**.
2. Change the **HTTP (80)** source from `0.0.0.0/0` to `YOUR_IP/32`. Find your IP by searching "what is my IP". Click **Save rules**.

Test

- From your home Wi-Fi, refresh Instance-A's page. It should **load**.

- From a mobile hotspot or other network, refresh it. It should **time out**.
- On Instance-B, run `curl http://<Public-IP-of-Instance-A>`. It will now **fail**, because Instance-B's public IP is not your home IP.

Change 2: Block traffic with a Network ACL

1. Go to **Subnets**, select the subnet that holds your instance, open the **Network ACL** tab, and click the NACL ID.
2. Open **Inbound rules → Edit inbound rules → Add new rule**.
3. Add a rule: **Rule number** `90`; **Type** HTTP (TCP 80) or All traffic; **Source** `YOUR_IP/32`; **Allow/Deny** **Deny**.
4. Click **Save changes**.

> **Why rule 90?** NACL rules are evaluated from the lowest number up and the first match wins. The default allow rule is number 100, so your deny rule must be lower than 100.

Test: From your home network, refresh Instance-A's page. It now fails even though SGA allows your IP. The NACL blocks traffic at the subnet boundary, before the security group is checked.

Remember: Security groups are stateful and only allow. NACLs are stateless and can allow or deny. Delete the deny rule when you finish testing.

Change 3: Limit SGB HTTP to SGA only

1. Go to **Security Groups → SGB → Edit inbound rules**.
2. Change the **HTTP (80)** source from `0.0.0.0/0` to **Custom → SGA** (select its security group ID). Click **Save rules**.

Test

- In a browser, open Instance-B's public IP. It should **time out**.
- On Instance-A, run `curl http://<Private-IP-of-Instance-B>`. It should **succeed**.

> **Important:** Use Instance-B's **private** IP for this test. A rule that references a security group only matches traffic from that group's private addresses. Using B's public IP sends the request out through the Internet Gateway and it will not match.

Quick Troubleshooting

Problem	Likely cause and fix
Browser shows nothing for the public IP	The URL uses `https://`, or Apache has not started. Use `http://` and check the User data ran.
No public IP on the instance	Auto-assign was off. Enable it on the subnet, or relaunch with Auto-assign public IP set to Enable.
Ping from A to B fails	Check that SGB has the ICMP rule with SGA as the source, and use B's private IP.
NACL deny does not work	Its rule number is higher than the allow rule. Use a lower number such as 90.
Cannot connect after Change 1	The source is not your current IP. Your home IP may have changed; update the rule.

Clean Up

To avoid charges, terminate both instances, then delete the VPC `lab1-custom` (this removes its subnets, route tables, and Internet Gateway). Delete SGA and SGB last, after the instances are gone.